RESEARCH SUMMARY

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1. Introduction

I am a geometric topologist studying dynamical structures in 3-manifolds and free-by-cyclic groups. These techniques, originally pioneered by Thurston in the 1970's, involve simplifying the dynamics by reducing the complexity—for example simplifying the dynamics of a flow on a hyperbolic 3-manifold (a continuous action $\mathbb{R} \curvearrowright M$) to the dynamics of $\pi_1 M$ acting on a circle (a discrete action $\pi_1 M \curvearrowright S^1$). In particular, these structures can usually (and conjecturally always) be studied via sphere filling curves $S^1 \to S^2$ which also encapsulate these dynamics. The rich theory of hyperbolic 3-manifolds fibering over the circle has also been generalized to free-by-cyclic groups—groups whose behavior is suspiciously similar to these 3-manifolds. The bulk of my work has been focused on two fronts:

(1) In work with my advisor Steven Frankel, we show any that sphere filling curve $S^1 \to S^2$ approximated by embeddings has the same rich structure as those coming from 3-manifold constructions. [FK26].
(2) In work with Maxwell Plummer, we organize the infinitely many distinct maps associated to different splittings of a free-by-cyclic group into finitely many such maps, generalizing a similar result in 3-manifolds. [KP26]

In the future, I plan to build on this work and:

(1) Use 1 to construct 3-manifolds from sphere-filling curves.
(2) Use 1 to construct new sphere-filling curves associated to dynamical structures which are not known to admit them.
(3) Compare the maps in produced in 2 to one another to better understand the various splittings of a free-by-cyclic group.
(4) Generalize the results of 2 to topological structures in these groups that do *not* come from group-theoretic splittings.

2. Background: Fibered 3-manifolds and Cannon-Thurston Maps

Suppose that $f : \Sigma \to \Sigma$ is a homeomorphism of a closed surface with genus at least two. Using f, we may create a 3-manifold as follows: First, observe that $\Sigma \times [0,1]$ is a compact 3-manifold with boundary given by two copies of Σ. We then create M_f, the ***mapping torus*** of f, by gluing one boundary component to the other along f:

$$M_f = \Sigma \times [0,1]/(p,0) \sim (f(p),1)$$

Such an M_f is a closed 3-manifold with a fibration $M_f \to S^1$ whose point-preimages are each a copy of Σ. Further, the dynamics of f determines the geometric structure of M_f. A celebrated theorem of Thurston is that M_f is hyperbolic (a geometric property) if and only

if f does not fix the isotopy class of any simple closed curve on Σ (a topological property) [Thu88]. Homeomorphisms with this property are called ***pseudo-Anosov***.

M_f comes with a ***suspension flow*** φ given by flowing in the 'vertical' direction which is transverse to the ***foliation by fibers***, the image of $\{\Sigma \times \{x\} \mid x \in [0,1]\}$ under the quotient map. We construct the ***orbit space*** of φ by lifting φ to a flow $\tilde{\varphi}$ on the universal cover $\tilde{M}$ and quotienting each flow line of $\tilde{\varphi}$ to a point: $\mathcal{O}_\varphi = \tilde{M}/\tilde{\varphi}$. The orbit space $\mathcal{O}_\varphi$ is the prototypical example of a ***bi-foliated plane***—a plane P supporting a pair of foliations (decompositions of P into lines and pronged singularities) which meet transversely at every point.

Giving Σ and M_f their natural hyperbolic structures, we may identify their universal covers with hyperbolic 2- and 3-space respectively: as such, the inclusion $\Sigma \hookrightarrow M_f$ lifts to a map $\mathbb{H}^2 \to \mathbb{H}^3$. A celebrated theorem of Cannon and Thurston [CT07] is that this map extends continuously to their natural compactifications $\mathbb{H}^2 \sqcup S^1_\infty \to \mathbb{H}^3 \sqcup S^2_\infty$. Restricting our attention to the boundaries, the ***Cannon-Thurston map*** $CT : S^1_\infty \to S^2_\infty$ is $\pi_1 M$ equivariant and is therefore forced to be surjective.

2.1. **Overview.** My research program can be broadly divided into two subfields and I devote a section below to each of them. "Cannon-Thurston Maps onto S^2" covers the work done in [FK26] in the setting of hyperbolic 3-manifolds. "Cannon-Thurston Maps for Free-by-Cyclic Groups" covers the work done in [KP26] generalizing well-known results in 3-manifolds to a new setting.

3. Cannon-Thurston Maps onto S^2

A flow ψ on a closed hyperbolic 3 manifold is a ***pseudo-Anosov flow*** if it is locally modeled on the suspension flow of a pseudo-Anosov map. The orbit space $\mathcal{O}_\psi$ is topologically a bifoliated plane whose foliations are determined by the dynamics of the flow [FM01]. In [Fen12], Fenly shows that $\mathcal{O}_\psi$ can be compactified into a closed disk $\mathcal{D}_\psi = \mathcal{O}_\psi \sqcup S^1_\psi$ and that there is a Cannon-Thurston map $S^1_\psi \to S^2_\infty$. On the other hand, a flow φ on a closed hyperbolic 3 manifold is a ***quasigeodesic flow*** if, after lifting φ to a flow $\tilde{\varphi}$ on $\tilde{M}$, each flowline of $\tilde{\varphi}$ stays a bounded distance from a geodesic; as such every flow line determines two points in S^2_∞. By work of Calegari, the orbit space $\mathcal{O}_\varphi$ of a quasigeodesic flow a plane which can be naturally compactified to a closed disk, $\mathcal{D}_\varphi = \mathcal{O}_\varphi \sqcup S^1_\varphi$ [Cal06], Frankel then shows the existence of a Cannon Thurston Map $S^1_\varphi \to S^2_\infty$ for φ [Fra15].

Recent exciting work of Frankel and Landry unifies these two pictures: If φ is a quaisgeodesic flow on M, then there is a closed, flow-invariant subset $M_L \subset M$ and a surjection $p : M_L \to M$ such that p semiconjugates the quasigeodesic flow on M_L to a pseudo-Anosov flow on M [FL25]. In particular, this is done *without* changing the Cannon-Thurston map. The key insight is that one can modify the flow space by considering the corresponding ***circle laminations***—a closed set of unordered pairs of points in S^1; the laminations provide data about how the Cannon-Thurston map identifies points on S^1_φ.

In its simplest form, a ***universal circle*** for a 3-manifold is a circle S^1_u and an action $\pi_1 M \curvearrowright S^1_u$ which contains information about a geometric or dynamical structure in M. The prototypical examples are the circle at the boundary of the orbit space S^1_φ of a quasigeodesic

pseudo-Anosov flow and the ***leftmost/rightmost circles*** $\mathcal{C}^{\ell/r}$ of an ***Anosov foliation***. Recent work by Buckminster [Buc26] produces a Cannon-Thurston map for $\mathcal{C}^{\ell/r}$, inspiring the following question:

Question 1. *Suppose that S^1_u is a universal circle for M. Does there exist a Cannon-Thurston map $\varphi : S^1_u \to S^2_\infty$?*

While universal circles arising from orbit spaces come with the additional data in the form of a pair of circle laminations and corresponding bi-foliated plane, this is not always the case—$\mathcal{C}^{\ell/r}$ lack this structure. Conjecturally, every universal circle is the boundary of the orbit space of some pseudo-Anosov flow [LMT24].

Question 2. *Suppose that S^1_u is a universal circle for M admitting a Cannon-Thurston map $\varphi : S^1_u \to S^2_\infty$. Does φ admit a pair of laminations extending to a bifoliated plane?*

My work with Frankel opens a new angle of attack for this question:

Theorem 3.1 (Frankel-K. [FK26])**.** *Suppose that $f_\infty \colon S^1 \to S^2$ is a sphere filling curve and that there is a continuous family $f_t : S^1 \to S^2$, $t \in [0, \infty]$, such that f_t is an embedding when $t < \infty$. Then f_∞ supports a pair of laminations $\Lambda^\pm$ extending to bifoliated plane.*

Thus, we have reduced Question 2 to showing that the Cannon-Thurston maps are approximated by embeddings. Further, the construction in Theorem 3.1 is natural in the sense that if $f_\infty : S^1 \to S^2$ is equivariant under the action of some group G, then the action of G in S^1 preserves the laminations $\Lambda^\pm$ and G acts on the corresponding bifoliated plane.

This work is a part of a larger program by Calegari as work towards ***Cannon's Conjecture***. Let M be a compact, metrizable space and $\Theta(M)$ the space of distinct triples of point in M. A group action $G \curvearrowright M$ is a ***uniform convergence group action*** if the induced action $G \curvearrowright \Theta(M)$ is properly discontinuous and co-compact. Combining results of Tukia, Gabai, Casson-Jungreis and Freden, [Tuk86; Gab92; CJ94; Fre95] we have the following characterization of uniform convergence group action on circles.

Theorem 3.2. *Suppose that G acts as a uniform convergence group action on S^1. Then there is a discrete group $\Gamma \subset \mathrm{Isom}(\mathbb{H}^2)$ such that $\Gamma \curvearrowright \mathbb{H}^2 \sqcup \partial_\infty \mathbb{H}^2$ is properly discontinuous and cocompact and the actions $G \curvearrowright S^1$ and $\Gamma \curvearrowright \partial_\infty \mathbb{H}^2$ are conjugate.*

A natural follow up question is to ask if there are similar statements for $G \curvearrowright S^n$ and $\Gamma \subset \mathrm{Isom}(\mathbb{H}^{n+1})$. If $n > 2$, the statement is false; Cannon's Conjecute is the case where $n = 2$:

Conjecture 3.3 (Cannon's Conjecture)**.** *Suppose that G acts as a uniform convergence group on S^2. Then G is conjugate to a discrete group $\Gamma \in \mathrm{Isom}(\mathbb{H}^3)$.*

If the group G is torsion free, then Cannon's Conjecture implies that G is $\pi_1 M$ for some closed, hyperbolic 3-manifold M. In their September release, OpenAI claims to have resolved Cannon's Conjecture positively, [Ope26], but their proof does not interface with any "internal structures" of the corrospondeng 3-manifold, an important part of Calegari's program.

Calegari and Loukidou introduce ***CaTherine Wheels*** and ***Zippers*** [CL24; CL26] as the start of a dictionary between structures on 3-manifolds and structures on S^2. CaTherine

wheels are a class of sphere filling curves $f\colon S^1 \to S^2$ such that for every closed interval $I \subset S^1$, $f(I)$ is a closed disk and $f(\partial I) \subset \partial f(I)$. All such maps are approximated by embeddings and Theorem 3.1 is a key step in developing their dictionary. Importantly however, there are many topological and dynamical structures whose Cannon-Thurston maps are *not* CaTherine Wheels. The proper generalizations—P-Zippers and P-CaTherine Wheels—are yet to be formalized and the ideas are rich for exploration. As such, a major future direction for my research is to build 3 manifolds from their Cannon-Thurston Maps:

Conjecture 3.4. *Suppose G acts as a uniform convergence group action and that $f\colon S^1 \to S^2$ is a G-invariant sphere filling curve approximated by embeddings. Then there is a closed, hyperbolic 3-manifold M and a quasigeodesic flow φ on M such that $G = \pi_1 M$ and f is the Cannon-Thurston map for φ.*

4. Cannon-Thurston Maps for Free-by-Cyclic Groups

A celebrated result of Mj (formerly Mitra) says that whenever

$$1 \to H \to G \to Q \to 1$$

is a short exact sequence of infinite word hyperbolic groups, there is a surjective Cannon-Thurston map $\partial_\infty H \to \partial_\infty G$ between the Gromov boundaries of H and G [Mit98]. Cannon and Thurston's result is then the case where $H = \pi_1\Sigma$, $G = \pi_1 M$, and $Q = \mathbb{Z}$; notice that we can write G as the semidirect product $\pi_1 M \simeq \pi_1\Sigma \rtimes_{f_*} \mathbb{Z}$ where $f_*\colon \pi_1\Sigma \to \pi_1\Sigma$ is induced by gluing map $f : \Sigma \to \Sigma$.

Let F_N denote the free group on N letters; given an outer automorphism $\varphi \in \mathrm{Out}(F_N)$, we may form the ***free-by-cyclic*** $G \simeq F_N \rtimes_\varphi \mathbb{Z}$. As long φ is fully irreducible and atoroidal (technical conditions on φ to ensure it is sufficiently mixing), G is a hyperbolic group that shares many properties with a fibered hyperbolic 3-manifold. By the aforementioned result of Mj, there is a Cannon-Thurston map $\partial_\infty F_N \to \partial_\infty G$. Unlike the fibered setting where these spaces at infinity are less well-behaved: if $N \geq 2$, then $\partial_\infty F_N$ is a Cantor set and $\partial_\infty G$ is the Menger Sponge [KK00].

While a hyperbolic free-by-cyclic group $G \simeq F_N \rtimes_\varphi \mathbb{Z}$ is a purely group theoretic construction, there are many ways to reintroduce topology. An ***expanding train track map*** for φ is a graph Γ with smoothing data around each vertex and a homotopy equivalence $f\colon \Gamma \to \Gamma$ inducing φ on $\pi_1\Gamma$. Dowall-Kapovich-Leininger use the data of an expanding train track map to construct the ***folding mapping torus***, a $K(G,1)$ with a fibration $X_f \to S^1$ whose point preimages are graphs homotopy equivalent to Γ and monodromy f [DKL15]. The folded mapping torus comes equipped with the ***suspension semi-flow*** ψ_t (an action $\mathbb{R}_+ \curvearrowright X_G$) defined analogously to the suspension flow on a mapping torus; as f is not invertible, we can only flow forwards, hence the restriction to a semi-flow.

The first natural question about a hyperbolic free-by-cyclic group is if the splitting is unique. In general, the answer is a resounding no—most free-by-cyclic groups split infinitely many ways with infinitely many different monodromies. The splittings of G are governed by the ***symmetric BNS invariant***, a finite union of open cones $\Sigma_s(G) \subset H^1(G;\mathbb{R})$ [BNS87]. A surjective homomorphism $u : G \to \mathbb{Z}$ satisfies $[u] \in \Sigma_s(G)$ if and only if $\ker u$ is finitely generated; in this case, $\ker u$ is free and we can write $G \simeq \ker u \rtimes \mathbb{Z}$ [GMSW01]. The folded mapping torus is an invariant of an entire component of the BNS invariant. If $[u]$ and $[v]$

belong to the same component of the BNS invariant with monodromies φ, ψ and expanding train track maps $f : \Gamma \to \Gamma$ and $g : \Theta \to \Theta$, then there is a homeomorphism $X_f \to X_g$ beteen their folded mapping tori sending semiflow lines to semiflow lines and inducing an isomorphism between the splittings $F_N \rtimes_\phi \mathbb{Z} \simeq G \simeq F_{N'} \rtimes_\psi \mathbb{Z}$. The utility of the BNS invariant motivates:

Question 3. *Given a splitting of a free-by-cyclic group $G \simeq F_N \rtimes_\varphi \mathbb{Z}$, what properties of the splitting depend only on the group G? Which properties depend only on the component of the BNS Invariant?*

Question 3 has many partial answers. For example, Mutanguha proved that (full) irreducibility of φ is an invariant of the group [Mut21], building on work of Dowdall–Kapovich–Leininger, who showed it was an invariant of a component of the BNS invariant [DKL17]. Maxwell Plummer and I answer Question 3 for the Cannon-Thurston Map.

Theorem 4.1 (K.-Plummer [KP26])**.** *Let $G \simeq F_N \rtimes_\varphi \mathbb{Z}$ be a hyperbolic free-by-cyclic group with fully irreducible monodromy. The Cannon-Thurston map $\partial_\infty F_N \to \partial G$ depends only on the component of the BNS Invariant.*

We construct a ***Universal Cantor*** set $\mathfrak{C}$ and a map $\mathcal{R} : \mathfrak{C} \to \partial_\infty G$ such that for any $u : G \to \mathbb{Z}$ which belongs to the same component as φ there is an identification $h_u : \ker u \to \mathfrak{C}$ such that the Cannon Thurston map $CT_u : \ker u \to \partial_\infty G$ factors as $CT_u = \mathcal{R} \circ h_u$.

Building on this result, I hope to compare the universal Cantor sets across different components of the BNS Invariant. In In the corrosponding question in 3-manifolds, universal circles from different components of the BNS Invariant for $\pi_1 M$ are distinct in both their $\pi_1 M$ actions and their Cannon-Thurston maps. Thus, I hope to prove

Conjecture 4.2. *Suppose that G is a hyperbolic free-by-cyclic group with fully irreducible monodromy and let $\mathfrak{C}$ and $\mathfrak{C}'$ denote the universal Cantor sets associated to distince component of $\Sigma_s(G)$. Then the actions $G \curvearrowright \mathfrak{C}$ and $G \curvearrowright \mathfrak{C}'$ are not conjugate.*

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